MakeItFrom.com Menu (ESC)

Home > Iron Alloy > Wrought Precipitation-Hardening Stainless Steel > S46500 Stainless Steel

H1025 Hardened S46500 Stainless Steel

H1025 hardened S46500 stainless steel is S46500 stainless steel in the hardened (H) condition. It has the second highest ductility compared to the other variants of S46500 stainless steel.

The graph bars on the material properties cards below compare H1025 hardened S46500 stainless steel to: wrought precipitation-hardening stainless steels (top), all iron alloys (middle), and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

Mechanical Properties

Brinell Hardness

460

Elastic (Young's, Tensile) Modulus

190 GPa 28 x 10^6 psi

Elongation at Break

13 %

Fatigue Strength

840 MPa 120 x 10^3 psi

Poisson's Ratio

0.28

Reduction in Area

50 %

Rockwell C Hardness

50

Shear Modulus

75 GPa 11 x 10^6 psi

Shear Strength

1000 MPa 140 x 10^3 psi

Tensile Strength: Ultimate (UTS)

1650 MPa 240 x 10^3 psi

Tensile Strength: Yield (Proof)

1510 MPa 220 x 10^3 psi

Thermal Properties

Latent Heat of Fusion

280 J/g

Maximum Temperature: Corrosion

680 °C 1260 °F

Maximum Temperature: Mechanical

780 °C 1430 °F

Melting Completion (Liquidus)

1450 °C 2650 °F

Melting Onset (Solidus)

1410 °C 2570 °F

Specific Heat Capacity

470 J/kg-K 0.11 BTU/lb-°F

Thermal Expansion

11 µm/m-K

Otherwise Unclassified Properties

Base Metal Price

15 % relative

Density

7.9 g/cm^3 490 lb/ft^3

Embodied Carbon

3.6 kg CO_2/kg material

Embodied Energy

51 MJ/kg 22 x 10^3 BTU/lb

Embodied Water

120 L/kg 14 gal/lb

Common Calculations

PREN (Pitting Resistance)

15

Resilience: Ultimate (Unit Rupture Work)

200 MJ/m^3

Resilience: Unit (Modulus of Resilience)

5860 kJ/m^3

Stiffness to Weight: Axial

14 points

Stiffness to Weight: Bending

25 points

Strength to Weight: Axial

58 points

Strength to Weight: Bending

39 points

Thermal Shock Resistance

57 points

Alloy Composition

Iron (Fe)	72.6 to 76.1
Chromium (Cr)	11 to 12.5
Nickel (Ni)	10.7 to 11.3
Titanium (Ti)	1.5 to 1.8
Molybdenum (Mo)	0.75 to 1.3
Manganese (Mn)	0 to 0.25
Silicon (Si)	0 to 0.25
Carbon (C)	0 to 0.020
Phosphorus (P)	0 to 0.015
Sulfur (S)	0 to 0.010
Nitrogen (N)	0 to 0.010

All values are % weight. Ranges represent what is permitted under applicable standards.

Followup Questions

How are the material properties defined?

How do I compare this exact variant to another material?

Further Reading

ASTM A564: Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes

Welding Metallurgy of Stainless Steels, Erich Folkhard et al., 2012

ASTM A959: Standard Guide for Specifying Harmonized Standard Grade Compositions for Wrought Stainless Steels

Properties and Selection: Irons, Steels and High Performance Alloys, ASM Handbook vol. 1, ASM International, 1993

Corrosion of Stainless Steels, A. John Sedriks, 1996

ASM Specialty Handbook: Stainless Steels, J. R. Davis (editor), 1994

Advances in Stainless Steels, Baldev Raj et al. (editors), 2010